MA∗508 Time Series

Exam-like exercises for L8

Solve the following exercises.

1. Determine whether the process x_t satisfying $x_t + 1.7x_{t-1} + 0.6x_{t-2} = \epsilon_t - 1.4\epsilon_{t-1} + 0.4\epsilon_{t-2}$, where ϵ_t is white noise and $t \in \mathbb{Z}$, is causal or not. Give a reason for your answer.

2. Determine whether the process x_t satisfying $x_t + 1.2x_{t-1} + 0.6x_{t-2} = \epsilon_t - 1.4\epsilon_{t-1} + 0.5\epsilon_{t-2}$, where ϵ_t is white noise and $t \in \mathbb{Z}$, is causal or not. Give a reason for your answer.

3. Determine whether the process x_t satisfying $x_t + 1.3x_{t-1} + 0.3x_{t-2} = \epsilon_t - 1.4\epsilon_{t-1} + 0.6\epsilon_{t-2}$, where ϵ_t is white noise and $t \in \mathbb{Z}$, is causal or not. Give a reason for your answer.

4. Determine whether the process x_t satisfying $x_t - 0.3x_{t-1} - 0.7x_{t-2} = \epsilon_t + 0.9\epsilon_{t-1} - 0.5\epsilon_{t-2}$, where ϵ_t is white noise and $t \in \mathbb{Z}$, is invertible or not. Give a reason for your answer.

5. Determine whether the process x_t satisfying $x_t - 0.3x_{t-1} + 0.7x_{t-2} = \epsilon_t + 0.5\epsilon_{t-1} - 0.4\epsilon_{t-2}$, where ϵ_t is white noise and $t \in \mathbb{Z}$, is invertible or not. Give a reason for your answer.

6. Determine whether the processes x_t from Exercises 1-5 are stationary or not-stationary.